

Lucas Goncalves

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SUMMARY

AI Engineer with 2+ years building NLP systems, LLM agents, and production ML pipelines using Python, TensorFlow, and LangChain. Automated mission-critical enterprise workflows processing millions in daily transactions at a global agritech manufacturer, reducing corporate process resolution time by ~50%. Architected FAPESP-sponsored RAG agent system (ReAct + LangGraph + Qdrant) as sole designer, indexing 500+ research documents with hybrid retrieval and hallucination detection. Seeking a full-time AI/ML Engineer role on a distributed remote team building production-grade intelligent systems at scale.

KEY ACHIEVEMENTS

- **Automated mission-critical after-sales workflow:** across national and international markets, shifting a global agritech manufacturer's team to review-only execution and reducing corporate process resolution time by ~50%.
- **Engineered ML-based soil texture classification mobile app:** awarded 3rd place among 1,300+ submissions at FETEPS 2025 and accepted for peer-reviewed poster presentation at ICPA 2026 / 17th ConBAP.
- **Built end-to-end NLP pipeline processing 1.6M tweets:** achieving 0.71 macro F1 on zero-shot RoBERTa baseline, with a Rust CLI preprocessing module delivering measurable latency reduction over pure-Python batch workloads at scale.

EXPERIENCE

Jacto | *After-Sales Intern*

Nov 2024 - Present

Global agricultural machinery manufacturer operating across 100+ countries with over 3,000 employees and industrial-scale after-sales operations.

- Redesigned mission-critical after-sales workflow by engineering a Python automation layer integrating Salesforce and SAP APIs, eliminating manual cross-system data entry and reducing corporate process resolution time by ~50%
- Identified 10+ usability improvements across QA cycles and implemented code logic changes that streamlined 3 manual workflow stages, reducing operational errors in cross-system data entry (Salesforce, SAP) and improving process execution speed by ~30%.
- Engineered automated data collection pipeline using Python (Requests and BeautifulSoup) with scheduled execution, replacing a recurring manual process and eliminating ~5 hours/week of error-prone database updates

PROJECTS

SB100 Agent System | github.com/LukeSantossz/sb100-agents | *FAPESP-Sponsored Research*

- Sole designer and developer of FAPESP-funded RAG agent system: architected ReAct-based agent over LangGraph state graph with hybrid retrieval pipeline (RRF + late interaction re-ranking) against Qdrant vector store indexing 500+ research documents. Implemented source traceability layer and hallucination detection module.

VisioSoil App | github.com/LukeSantossz/visiosoil-app | *3rd Place at 16th FETEPS*

- Building offline-first mobile app for soil texture classification using transfer learning on GCP and on-device SLM agent delivering GPS-contextualized agronomic recommendations. Ranked top 250 of 1,300+ at FETEPS 2025; accepted for poster presentation at ICPA 2026 / 17th ConBAP with abstract under peer review.

Tweet Sentiment Analysis | github.com/LukeSantossz/tweet-sentiment-analysis

- Engineered high-performance Rust CLI for preprocessing delivering measurable latency reduction over Python at batch scale on 1.6M tweets. Built full NLP pipeline with validated EDA, preprocessing module (6 cleaning functions, 11 tests, tokenization at 99th percentile), and zero-shot RoBERTa baseline achieving 0.71 macro F1.

TECHNICAL SKILLS

- **Languages:** Python, Rust, Dart, JavaScript
- **Frameworks & Libs:** TensorFlow, Keras, Scikit-learn, HuggingFace, OpenCV, LangChain, FastAPI, Flutter
- **Databases:** PostgreSQL, MongoDB, Qdrant, Firestore
- **Cloud & Infra:** GCP, Vertex AI, AWS, Docker, GitHub Actions, Google Earth Engine

EDUCATION

FATEC, Sao Paulo State Technology College

Jan 2024 - Dec 2026

B.Tech, Big Data in Agribusiness

- **Coursework:** Machine Learning, Artificial Intelligence, Data Structures, Databases (SQL & NoSQL), API & Microservices, Cloud Architectures, Statistics